

Chemistry Questions Waec Candidate/Private

Question 1

(a) State

(i) Pauli's Exclusion principle

(ii) Hund's rule of maximum multiplicity. [4 marks]

(b) Write the electronic configuration of each of the following ions of copper:

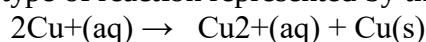
I. Cu^+

II. Cu^{2+} .

[29Cu]

(ii) Give the number of unpaired electrons in each of the ions in 1(b) (i) above.

(iii) State the type of reaction represented by the following equation:



(iv) Write the formula of one compound of Cu^+ . [6 marks]

(c) (i) Name the type of radiation that will

I. penetrate lead block

II. be stopped by thin paper.

(ii) Give the charge on each of the radiations mentioned in 1(c) (i) above.

(iii) What term is used to describe each of the following nuclear processes?

I. Combination of two lighter nuclei to form a heavy nucleus

II. Splitting of a heavy nucleus into two or more lighter nuclei

III. Time required for one-half of the atoms of a radioactive substance to decay [7marks]

(d) Arrange the following ions in order of increasing size. Give a reason for your answer in each case.

I. Li^+ , K^+ , Na^+

II. O^{2-} , F^- , N^{3-} . [4 marks]

(e) Determine the percentage composition of phosphorus and oxygen in phosphorus (V) oxide.

$$[\text{P} = 31, \text{O} = 16] \quad [4 \text{ marks}]$$

Question 2

(a) (i) Define in terms of electron transfer

I. oxidizing agent;

II. reducing agent.

(ii) Write a balanced equation to show that carbon is a reducing agent.

(iii) State the change in oxidation number of the specie that reacted with carbon in 2 (a)(ii). [5 marks]

(b) A gas X has a vapour density of 32. It reacts with sodium hydroxide solution to form salt and water only. It decolourizes acidified potassium tetraoxomanganate (VII) solution and reacts with H_2S to form sulphur. Using the information provided:

identify gas X;

state two properties exhibited by X;

give two uses of X. [5 marks]

(c) Consider the following substances:

- (i) sodium;
- (ii) lead (II) iodide;
- (iii) hydrogen;
- (iv) magnesium;
- (v) oxygen.

Which of the substances

- (i) conducts electricity?
- (ii) is produced at the cathode during electrolysis of $\text{H}_2\text{SO}_4(\text{aq})$?
- (iii) corresponds to the molecular formula AB_2 ?
- (iv) is an alkaline earth metal? [5 marks]

- (d) (i) Define the term salt.
- (ii) Mention two types of salt.
- (iii) Give an example of each of the salts mentioned in 2(d)(ii) above. [6 marks]

(e) In a neutralization reaction, dilute tetraoxosulphate (VI) acid completely reacted with sodium hydroxide solution.

- (i) Write a balanced equation for the reaction.
- (ii) How many moles of sodium hydroxide would be required for the complete neutralization of 0.50 moles of tetraoxosulphate (VI) acid? [4 marks]

Question 3

- (a) Consider the following reaction sequence.

- (i) What process leads to the formation of K?
- (ii) Write the formula of K.
- (iii) Write the structural formula of L and name L.
- (iv) Name An.
- (v) Write the structure of M and name M. [7 marks]

- (b) (i) What are carbohydrates?

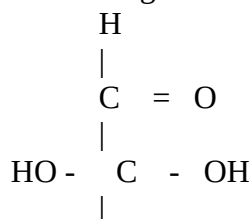
- (ii) Give one example each of a

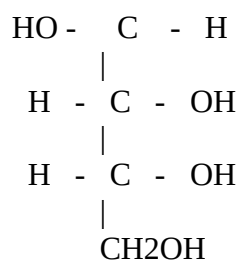
monosaccharide;

disaccharide;

polysaccharide. [5 marks]

- (c) Consider the following structure of a simple sugar.





- (i) Which functional group makes the compound a reducing agent?
- (ii) State what would be observed when
 (I) the compound is mixed with Fehling's solution and boiled;
 (II) few drops of concentrated H_2SO_4 is added to the sample of the compound.
- (iii) Write an equation for the reaction in 3(c)(ii)(II). [6marks]
- (d) A hydrocarbon Z with molecular mass 78 on combustion gave 3.385 g of CO_2 and 0.692 g of H_2O . Determine the molecular formula of Z.
 [H = 1, C = 12, O = 16] [7 marks]

Question 4

- (a) (i) Define covalent bond.
 (ii) Give two properties of covalent compounds
 (iii) With the aid of a diagram, show how ammonia molecule is formed.
 (iv) Illustrate with a diagram the formation of ammonium ion.
 (v) What type of bond(s) exist(s) in ammonia,
 ammonium ion? [1H, 7 N] [10 marks]
- (b) (i) Write three subatomic particles with their corresponding relative masses.
 (ii) Name the possible states in which water can exist. [9 marks]
- (c) (i) State Graham's law of diffusion.
 (ii) Arrange the following gases, He, CH_4 and N_2 in order of increasing rates of diffusion. Give a reason for the order.
- [H = 1, He = 4, C = 12, N = 14] [4 marks]
- (d) Draw the structures of the following compounds:
 (i) 2, 3-dimethylbutane;
 (ii) 1, 4-dibromocyclohexane. [2 marks]

Question 7

- (a) (i) What are acidic oxides?
 (ii) Give one example of each of the following oxides:

- I. acidic oxide;
- II. basic oxide;
- (III) amphoteric oxide;
- (III) neutral oxide. [5 marks]

(b) (i) Define each of the following terms:

- (I) Heat of combustion;
- (II) Heat of neutralization.

(ii) Write an equation to illustrate each of the terms in 7(b)(i) above.

(iii) Given that the standard heat of combustion of butane (C₄H₁₀) is + 5877 kJmol⁻¹, calculate the heat of combustion of 14.5 g butane.

[H = 1, C = 12] [10 marks]

(c) (i) Name two allotropes of sulphur.

(ii) State one difference between the two allotropes. [3 marks]

(d) (i) Give two characteristics of noble gases.

(ii) State one use each of

(I) He;

(II) Ar. [4 marks]

(e) State what is observed on warming ammonium trioxonitrate (V) with sodiumhydroxide. [2 marks]

Question 8

(a) (i) Define hard water.

(ii) Name two substances responsible for hardness in water.

(iii) Give two methods for the removal of hardness in water. [6 marks]

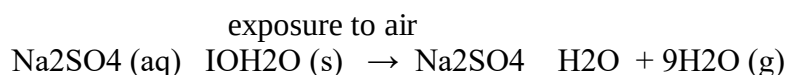
(b) (i) What are the raw materials required for the manufacture of tetraoxosulphate (VI) acid by the contact process?

(ii) Write an equation for the reaction that requires a catalyst in the contact process.

(iii) State the catalyst used in 8(b)(ii). [5 marks]

(c) (i) Give two uses of sodium tetraoxosulphate (VI)

Consider the reaction represented by the following equation:



(ii) What name is given to this type of reaction?

(iii) Calculate the solubility of Na₂CO₃ at 25°C, if 30.0cm³ of its saturated solution at that temperature gave 1.80 g of the anhydrous salt.

[C = 12, O = 16, Na = 23] [7 marks]

(d) (i) Define the term activated complex.

- (ii) State one reason why a collision may not produce a chemical reaction.
 (iii) The formation of water gas is represented by the following equation.



Draw an energy profile diagram for the reaction showing the

I. activated complex

II. enthalpy of reactants. [7 marks]

Question 9

(a)

Define the term hardness of water.

Give two advantages of hard water.

Give two disadvantages of hard water.

State how each of the following procedures can be carried out in the laboratory:

Removal of sediment in a sample of water .

Softening temporarily hard water without heating it. [8 marks]

(b)

(i) Write an equation for the laboratory preparation of chlorine.

(ii) Name the products formed when chlorine reacts with hot concentrated sodium hydroxide solution.

(iii) What is observed when moist blue litmus paper comes in contact with chlorine?

[6 marks]

(c)

(i) Give two properties of copper which make it useful for ornamental work.

(ii) Give two uses of copper (II) tetraoxosulphate (VI).

(iii) Name one alloy of:

copper .

aluminium.

(iv) State the reason why metals are good conductors of electricity.

[7 marks]

(d)

(i) Give two uses of trioxonitrate (V) acid.

(ii) In the preparation of trioxonitrate (V) acid in the laboratory from trioxonitrate (V) salts, state:

what other reagent is required?
the reason why an all-glass apparatus must be used.

[4 marks]

Question 10

Name one allotrope of oxygen.
Draw a diagram for the laboratory preparation of dry oxygen gas from potassium trioxochlorate (V).

[6 marks]

Ammonia is manufactured in the industry by the Haber process.
State the source of the raw materials.
Name two impurities removed during the purification of the materials.
In what state is the final product collected.
Give a reason for the answer given in 4(b)(iii).

[6 marks]

Name one solid which reacts with dilute hydrochloric acid to produce sulphur (IV) oxide.

Describe a test for sulphur (IV) oxide.
State how sulphur (IV) oxide behaves in 4(c)(ii).
In the production of tetraoxosulphate (VI) acid, name two sources of sulphur.

[7 marks]

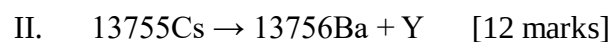
- (i) State three major chemical industries in Nigeria.
- (ii) List the raw material(s) used in each of the industries stated in 4(d)(i).

[6 marks]

question 11

- (a) (i) Give the name and nature of the radiations that are emitted during radioactivity.
- (ii) State two differences between chemical reaction and nuclear reaction.
- (iii) Balance the following nuclear reactions and identify X and Y





(b) The electron configuration of an element X is:

1s²2s²2p⁶3s²3p⁵

- (i) Deduce the atomic number of X.
- (ii) To what group does X belong?
- (iii) Give two properties of the group to which the element X belong.
- (iv) Identify element X by name.
- (v) Write a balanced equation to represent the reaction between the element X and hot concentrated NaOH

(c) (i) Explain why:

1. graphite is used as a lubricant .

II. diamond is used as an industrial cutting tool.[7 marks]

- (ii) Write an equation to represent the reaction between aqueous BCL and solution.
- (iii) Name the type of reaction represented by the equation. [6 marks]

Question 1 2

(a) (i) Define saturated solution.

(ii)The solubility of KN03 at 20°C was 3.00 mol dm⁻³ If 67.0g of KN03 was added to 250cm³ of water and stirred at 20°C, determine whether the solution formed was saturated or not at that temperature.

[7 marks]

(b) (i) Distinguish between dative bond and covalent bond.

(ii) Explain why sugar and common salt do not conduct electricity in the solid state.

(iii) State the type of intermolecular forces present in:

1. hydrogen fluoride .

II. argon.

(iv) Consider the compounds with the following structures:

S - H ----N and O - H -----N

In which of the compounds is the hydrogen bond stronger? Give reason for your answer.

[12 marks]

(c) (i) State Dalton's Law of Partial Pressure.

(ii) If 200cm³ of carbon(IV) oxide were collected over water at 18°C and 700 mmHg, determine the volume of the dry gas at s.t.p.

[standard vapour pressure of water at 18°C = 15 mmHg] [6 marks]

question 13

(a) (i) Define a base according to Arrhenius concept.

(ii) Give one example of an Arrhenius base.

(iii) Identify each of the following substances in aqueous solutions as strong electrolyte,

non-electrolyte or weak electrolyte.

I. $\text{Cl}_2\text{H}_2\text{O}_{11}$.

II. NH_3

III. NaOH

(iv) Write a balanced equation to represent the reaction between CH_3COOH and KOH

[8 marks]

(b) Calculate the volume of $0.500 \text{ mol dm}^{-3}$ HCl required to-neutralize 20.00 cm^3 of $0.300 \text{ mol dm}^{-3}$ NaOH. [3 marks]

(c) Give the IUPAC name of each of the following salts:

(ii) NaOCl .

(iii) $\text{Mg}(\text{HCO}_3)_2$ [2 marks]

(d) (i) Define the term standard solution

(ii) Consider the following compounds:

NaOH and Na_2CO_3

Which of the compounds is suitable for the preparation of a standard alkaline solution? Give reason for your answer.

(iii) Fe completely reacted with dilute HCl.

1. Write an equation for the reaction.

II. If 3.08 g of Fe completely reacted with 50.0 cm^3 of 2.20 mol dm^{-3} HCl, calculate the relative atomic mass of the metal. [12 marks]

question 14

(a) (i) Give the two reasons why soda lime is used instead of caustic soda in the preparation of methane.

(ii) List two physical properties of methane.

(Hi) A hydrocarbon with a vapour of 29 contains 82.76% carbon and 17.24% hydrogen.

Determine the:

- I. empirical formula .
- ii. molecular formula of the hydrocarbon.

[H = 1.00 C = 12.00] [12 marks]

(b) (i) What is meant by the term isomerism?

(ii) Draw the structures of the two isomers of the compound with the molecular formula C_2H_6O .

(iii) Give the name of each of the isomers in 4(b)(ii).

(iv) State the major difference between the isomers. [7 marks]

(c) Give three deductions that could be made from the qualitative and quantitative analysis of a given organic compound. [3 marks]

(d) Give one chemical test to distinguish between propene and propane. [3 marks]

question 15

(a) Name one product of destructive distillation of coal that is

(i) Solid .

(ii) Liquid .

(iii) Gas. [3 marks]

(b) (i) What is the major component of synthetic gas?

(ii)

Give one reason why synthetic gas is not a major source of air pollution.

[2 marks]

(c) (i) Write a balanced chemical equation for the complete combustion of carbon.

(ii) State two:

I. physical .

II. chemical properties of the product in 7 (c)(i) [5 marks]

(d) (i) Name two allotropes of carbon that are

1. crystalline .

II. amorphous.

(ii) State one use of each of the allotropes named in 7(d)(i). [8 marks]

(e) (i) By means of balanced chemical equations only, outline the process of manufacture of H₂SO₄ by contact process.

(ii) State the function of H₂SO₄ in each of the following reaction equations:

1. $\text{C}_2\text{H}_5\text{OH}(\text{l}) \xrightarrow{\text{conc H}_2\text{SO}_4} \text{C}_2\text{H}_4(\text{g})$.

II. $\text{Pb}(\text{NO}_3)_2(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{PbSO}_4(\text{s}) + 2\text{HNO}_3(\text{aq})$ [7 marks]

question 16

(a) (i) Explain why water is referred to as a universal solvent.

(ii) Give one chemical test for water. [5 marks]

(b) A current of 1.25A was passed through an electrolytic cell containing dil. H₂SO₄ for 40 minutes

(i) Write a balanced equation for the:

I. oxidation half reaction .

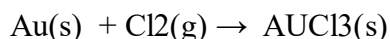
II. reduction half reaction .

III. overall reaction.

- (ii) Calculate the volume of gas produced at the anode at s.t.p.

[IF = 96500 C, Molar Volume of gas at s.t.p = 22.4 dm³ mol⁻¹] [10 marks]

- (c) Consider the reaction represented by the following equation:



- (i) Balance the equation
- (ii) If 1.250g of Au and 1.744g of Cl₂ were mixed,
- I. determine which of the reactants is in excess .
- II. calculate the excess amount,

[Au = 197.0, Cl = 35.5] [10 marks]

Question 17

- a. State **two** applications of titration in the laboratory. [2 marks]
- b. Write the reduction half-reaction equation to illustrate the reaction of acidified tetraoxomanganate (VII) ions with a reducing agent. [2 marks]
- c. Explain briefly why there is a general increase in the first ionization energies of the elements across the period in the periodic table. [3 marks]
- d. (i) Name **two** greenhouse gases.
- (ii) State **one** effect of an increased level of greenhouse gases on the environment. [3 marks]
- e. (i) Define *standard electrode potential*.
- (ii) State **one** factor that could affect the standard electrode potential. [3 marks]
- f. (i) State **two** properties of gases that determine their method of collection during preparation in the laboratory.
- (ii) State the method of collection of CO₂ gas. [3 marks]
- g. Name the reagents that could be used to test for the presence of tetraoxosulphate(VI) ions in aqueous solution. [2 marks]
- h. Distinguish between *hydrolysis* and *hydration*. [2 marks]
- i. State **three** sources of error during the titration of dilute HCl and aqueous NaOH. [3 marks]
- j. State the *Charles's law*. [2 marks]

questions 18

- a. An electrochemical cell is constructed with copper and silver electrodes.
- i. State the type of reaction occurring at **each** electrode.
- ii. Write a balanced equation for the overall cell reaction.

iii. Draw and label a diagram for the cell.
[9 marks]

b.i. Explain **briefly** the term *cracking*.

ii. State **two** conditions under which cracking takes place.

iii. What is the importance of cracking in the petroleum industry?

[5 marks]

c.i. Write the electron configuration for: 1. ${}^{29}\text{Cu}$; 2. ${}^{29}\text{Cu}^{+}$.

ii. In what way is the electron configuration of copper different from the general trend across the first transition element?

[5 marks]

d.i. Define the term *oxidation number*.

ii. Propan-1-ol can be oxidized to propanoic acid. 1. State the reagent necessary for this reaction to occur.

2. In an experiment, 5.67 g of propan-1-ol was oxidized to the acid with a 64% yield. Calculate the mass of propanoic acid produced.

[Molar mass: $\text{C}_2\text{H}_5\text{CH}_2\text{OH} = 60$; $\text{C}_2\text{H}_5\text{COOH} = 74$]

[6 marks]

question 19

a. An impure sample of zinc metal weighing 1.70 g reacted completely with 25.0 cm³ of 1.50 mol dm⁻³ hydrochloric acid.

i. Write a balanced chemical equation for the reaction.

ii. Determine the: 1. mass of pure zinc;

2. percentage purity of the sample. [Zn = 65.0]

[8 marks]

b. A white curdy precipitate of silver chloride is obtained by mixing aqueous solutions of sodium chloride and silver trioxonitrate (V). State what would happen when a portion of the precipitate is:

i. exposed to light;

ii. shaken with excess aqueous ammonia solution;

iii. shaken with dilute trioxonitrate (V) acid.

[4 marks]

c. Consider the following table:

Substance X	Substance Y	Substance Z
No fixed shape	Fixed shape	No fixed shape
No fixed volume	Fixed volume	Fixed volume
Low boiling point	High melting point	Low boiling point
Insoluble in water	Soluble in water	Miscible in water

State the substance which:

i. has the **strongest** binding force;

ii. is likely to have hydrogen bonding;

iii. could be a solvent for grease;

iv. is very compressible;

v. requires heating to melt.

[5 marks]

d.

i. Name the process involved in the conversion of a gas into its liquid state.

ii. Using the kinetic theory, explain **briefly** how a given volume of a gas could be converted to its liquid state.

[4 marks]

e.

i. Define the basicity of an acid.

ii. Determine the basicity of each of the following acids: 1. H_2PO_4^- ;

2. CH_3COOH .

[4 marks]

question 20

Question 5

(a)

i. Define the term *hardness of water*.

ii. Give **two** advantages of hard water.

iii. Give **two** disadvantages of hard water.

iv. State how each of the following procedures can be carried out in the laboratory:

1. Removal of sediment in a sample of water;

2. Softening temporarily hard water without heating it.

[8 marks]

(b)

i. Write an equation for the laboratory preparation of chlorine.

ii. Name the products formed when chlorine reacts with hot concentrated sodium hydroxide solution.

iii. What is observed when moist blue litmus paper comes in contact with chlorine?

[6 marks]

(c)

i. Give **two** properties of copper which make it useful for ornamental work.

ii. Give **two** uses of copper (II) tetraoxosulphate (VI).

iii. Name **one** alloy of: 1. Copper 2. aluminium.

iv. State the reason why metals are good conductors of electricity.

[7 marks]

(d)i. Give **two** uses of trioxonitrate (V) acid.

- ii. In the preparation of trioxonitrate (V) acid in the laboratory from trioxonitrate (V) salts, state: 1. what other reagent is required;
2. the reason why an all-glass apparatus must be used.
[4 marks]

question 21

(a) State

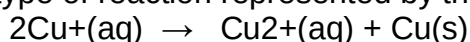
- (i) *Pauli's Exclusion principle*;
(ii) *Hund's rule of maximum multiplicity*. [4 marks]

(b) Write the electronic configuration of each of the following ions of copper:

- I. Cu^+ ;
II. Cu^{2+} .
[29Cu]

(ii) Give the number of unpaired electrons in each of the ions in 1(b) (i) above.

(iii) State the type of reaction represented by the following equation:



(iv) Write the formula of one compound of Cu^+ . [6 marks]

(c) (i) Name the type of radiation that will

- I. penetrate lead block;
II. be stopped by thin paper.

(ii) Give the charge on each of the radiations mentioned in 1(c) (i) above.

(iii) What term is used to describe each of the following nuclear processes?

- I. Combination of two lighter nuclei to form a heavy nucleus
II. Splitting of a heavy nucleus into two or more lighter nuclei
III. Time required for one-half of the atoms of a radioactive substance to decay [7marks]

(d) Arrange the following ions in order of increasing size. Give a reason for your answer in each case.

- I. Li^+ , K^+ , Na^+ ;
II. O^{2-} , F^- , N^{3-} . [4 marks]

(e) Determine the percentage composition of phosphorus and oxygen in phosphorus (V) oxide.

$$[\text{P} = 31, \text{O} = 16] \quad [4 \text{ marks}]$$

Question 21

(a)i. Define in terms of electron transfer:

1. *oxidizing agent*;
2. *reducing agent*.

ii. Write a balanced equation to show that carbon is a reducing agent.

iii. State the change in oxidation number of the specie that reacted with carbon in (a)(ii).
[5 marks]

(b) A gas **X** has a vapour density of 32. It reacts with sodium hydroxide solution to form salt and water only. It decolourizes acidified potassium tetraoxomanganate (VII) solution and reacts with H_2S to form sulphur. Using the information provided:

- Identify gas **X**;
 - State two properties exhibited by **X**;
 - Give two uses of **X**.
- [5 marks]

(c) Consider the following substances:

- sodium;
- lead (II) iodide;
- hydrogen;
- magnesium;
- oxygen.

Which of the substances:

- conducts electricity?
 - is produced at the cathode during electrolysis of $\text{H}_2\text{SO}_4(\text{aq})$?
 - corresponds to the molecular formula AB_2 ?
 - is an alkaline earth metal?
- [5 marks]

(d)

- Define the term *salt*.
 - Mention **two** types of salt.
 - Give an example of **each** of the salts mentioned in (d)(ii) above.
- [6 marks]

(e) In a neutralization reaction, dilute tetraoxosulphate (VI) acid completely reacted with sodium hydroxide solution.

- Write a balanced equation for the reaction.
 - How many moles of sodium hydroxide would be required for the complete neutralization of 0.50 moles of tetraoxosulphate (VI) acid?
- [4 marks]

question 23

(a) Consider the electron configuration of the following elements:

Q: $1s^2 2s^2 2p^1$

R: $1s^2 2s^2 2p^6 3s^2$

S: $1s^2 2s^2 2p^6 3s^2 3p^1$

T: $1s^2 2s^2 2p^6$

U: $1s^2 2s^2 2p^6 3s^2 3p^5$

- Which element(s) belong(s) to the same group?
- Which element(s) belong(s) to the same period?

3. Which element(s) ionize(s) by readily:
 - (i) Losing electrons?
 - (ii) Gaining electrons?
4. Which element(s) is/are noble gas(es)?
5. Which element(s) would react to form electrovalent compounds?

[8 marks]

(b)

1. (i) Define electronegativity.
2. (ii) State the trend of electronegativity on the periodic table.
3. (iii) Explain briefly why atomic radius increases down the group but decreases across the period in the periodic table.

[7 marks]

(c) List three characteristic properties of transition metals.

[3 marks]

(d)

1. If 15 g of potassium trioxochlorate (V) is heated in the presence of MnO_2 :
 - (i) Write a balanced equation for the reaction.
 - (ii) State the role of MnO_2 in the reaction.
 - (iii) Calculate the mass of chloride produced.

[7 marks]

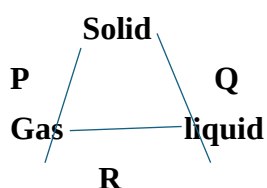
Questions 24

(a)

1. State Boyle's law.
2. Give the mathematical expression of Boyle's law.
3. Give **two** reasons why real gases deviate from ideal gas behavior.
4. State:
 - (i) Three postulates of the kinetic theory of gases.
 - (ii) Dalton's law of partial pressures.

(b)

1. (i) Consider the figure as illustrated below:



State the processes represented by **P**, **Q**, and **R** respectively.

2. (ii) Some solids when heated change directly to the gaseous state.

- (i) What name is given to this phenomenon?
- (ii) Name two substances which exhibit the phenomenon referred to in 2(b)(ii)(i).

(c)

1. If 80 cm³ of methane (CH₄) diffuses through a porous membrane in 20 seconds while 50 cm³ of gas **Q** diffuses through the same membrane in 25 seconds, calculate the molecular mass of gas **Q**. [CH₄ = 16.0]
2. Explain briefly why at room temperature fluorine is a gas while bromine is a liquid.

Question 25

(a)

1. What is an acid-base indicator?
2. When is an indicator said to be most suitable for an acid-base titration?
3. Consider the acid-base titration reaction represented by the following equation:
 - $\text{NaOH(aq)} + \text{CH}_3\text{COOH(aq)} \rightarrow \text{CH}_3\text{COONa(aq)} + \text{H}_2\text{O(l)}$
 - (i) Name a suitable indicator for the reaction.
 - (ii) Explain briefly your answer in 3(a)(ii)(i).

(b)

1. (i) Write the ionic equation for the reaction between zinc and silver trioxonitrate (V) solution.
 - (ii) Which type of reaction occurs between zinc and silver trioxonitrate (V) solution?
 - Give a reason for your answer.

(c)

1. (i) Consider the reaction represented by the following equation:
 - $\text{CaCO}_3\text{(s)} + 2\text{HCl(aq)} \rightarrow \text{CaCl}_2\text{(aq)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$
 - State **three** factors that could increase the rate of the reaction.
2. Explain briefly the observation that an increase in temperature generally increases the rate of reaction.

(d)

1. (i) What is solubility?
 - (ii) Distinguish between saturated solution and unsaturated solution.
 - A saturated solution of volume 10 cm³ yielded 0.06 g of its dry salt at 25°C. Calculate the solubility of the salt in g/dm³

Question 26

1. (i) Define hybridization.
 - (ii) State the type(s) of hybridization exhibited by carbon in propene.
 - (iii) Mention the shape of the:

- s-orbital;
 - p-orbital.
2. Diamond and graphite are both allotropes of carbon. Diamond is hard while graphite is soft. Explain briefly this observation.
 3. A monochloroalkane was analyzed by converting all the chlorine into chloride ions and precipitating the chloride as silver chloride. If 0.08226g of the chloroalkane gave 0.1837g of silver chloride:
 - (i) Calculate the molecular mass of the chloroalkane.
 - (ii) Determine the molecular formula of the chloroalkane.
 - (iii) Draw the structure of the chloroalkane and name it.

[H = 1.00; C = 12.0; Cl = 35.5; Ag = 108.0]

Questions 27

1. (a) (i) Sodium hydroxide can be prepared by the electrolysis of a concentrated solution of sodium chloride. Write the equation at the:
 - I. Cathode;
 - II. Anode.
 - (ii) Explain briefly why the electrolysis of molten alumina is considered environmentally friendly while that of molten sodium chloride is not.
2. (b) (i) Describe briefly the electrolysis of copper (II) tetraoxosulphate (VI) solution using copper electrodes.
 - (ii) State the color of the electrolyte in 7(b)(i):
 - Before the electrolysis;
 - After the electrolysis.
 - (iii) Give a reason for your answer in 7(b)(ii).
3. (c) (i) Name the impurities present in bauxite.
 - (ii) State how the impurities in bauxite can be removed.
4. (d) An aluminum piece of mass 3.14g reacted with hydrochloric acid at s.t.p.
 - (i) Write a balanced equation for the reaction.
 - (ii) Calculate the:
 - Mass of hydrogen produced;
 - Volume of hydrogen produced at s.t.p.

[H = 1.00; Al = 27.0; Molar volume = 22.4 dm³]

Question 28

- (a) Potassium, hydrogen gas, and potassium hydride exhibit different types of bonds.

Substance	Type of Bonding	Units which make up this substance

K		
H ₂		
KH		

- ☐ (b) (i) Arrange the following compounds in order of increasing boiling points:
H₂O, CH₄, H₂S.
- (ii) Give reasons for your answer in 8(b)(i).
- ☐ (c) (i) State two processes where the concept of oxidation number is applied.
- (ii) Write the formula of the oxide in which nitrogen exhibits an oxidation number of:
 - +1;
 - +2;
 - +3.
 - (iii) Differentiate between a drying agent and a dehydrating agent. Give an example of each agent.
- ☐ (d) (i) State two chemical properties of an acid.
- (ii) Classify each of the following solutions as acidic, basic, or neutral:
 - NH₄Cl(aq);
 - KCO₃aq);
 - Na₂SO₃aq);
 - CH₃COONa(aq).

Question 29

- ☐ (a) Define the term **hydrocarbon**. [2 marks]
- ☐ (b) State the conditions necessary for cracking long-chain hydrocarbons to produce more gasoline. [2 marks]
- ☐ (c) Define **oxidation** in terms of electron transfer. [2 marks]
- ☐ (d) State **two** reasons for electroplating metallic objects. [2 marks]
- ☐ (e) Give the names of the following compounds:
- (i) CH₃CH=CHCH₂CH₃
 - (ii) CH₃CH₂CH₂-C-OH. [2 marks]
- ☐ (f) State **Dalton's law of partial pressures**. [2 marks]
- ☐ (g) (i) Give **two** reasons for preferring alloys to their pure metals. [2 marks]
- (ii) List the constituent elements in steel. [5 marks]
- ☐ (h) State the type of bonding in the following substances:
- (i) aluminum chloride;
 - (ii) sodium metal;

- (iii) sodium chloride. [3 marks]
- (i) (i) What are **isomers**? [3 marks]
- (ii) State why helium atom does not react with other elements to form compounds. [3 marks]

Questions 30

(a) (i) Copy and complete the following table:

Element	Number of Neutrons	Electron Configuration	Group
39X			1
19Y		1s ² 2s ² 2p ⁵	
20Z			

1. State the family to which **each** of the elements belongs.
 2. Name the type of chemical bond that can exist between **X** and **Y**.
 3. State two properties of compounds which have the type of bond named in 2(a)(iii). [13 marks]
- (b) (i) Explain **briefly** why the reactivity of Group VII elements decreases down the group.
- (ii) Arrange the following species in order of decreasing size: 20Ca²⁺, 17Cl⁻, 19K⁺.
1. Give reasons for your answer in 2(b)(ii). [7 marks]
- (c) (i) Define the term **relative atomic mass**.
- (ii) Calculate the vapour density of a diatomic gas with the relative atomic mass 32. [5 marks]

Question 31

(a) Consider the following compounds:



1. State the family to which **each** organic compound belongs.
2. Give the IUPAC name of **each** compound.
3. Arrange the compounds in order of increasing boiling point.

Give reasons for your answer. [15 marks]

(b) Chloroethane can be prepared by reacting chlorine with ethane.

1. State the experimental condition required for this reaction to occur.
2. Write the equation for the initiation step.
3. Write equations for the propagation steps of the reaction.
4. Name the product formed at the termination step of the reaction. [5 marks]

(c) Ethanol is produced on a large scale by heating ethane with steam.

1. State **two** other conditions necessary for the reaction.
2. Write the balanced chemical equation for the reaction.
3. State **one** advantage of this method over the fermentation method. [5 marks]

Question 32

- (i) Define the term *allotropy*.
(ii) State the type of bond present in diamond and graphite.
- Explain **briefly** why:
 - diamond is hard while graphite is soft;
 - graphite is a good conductor of electricity while diamond is not. [10 marks]

(b) Under **two** different conditions carbon burns in air to form two gaseous products.

- State the condition under which **each** product is formed.
- Write a balanced equation for the formation of **each** product formed in 4(b)(i).
- Name the product formed under **each** condition stated in 4(b)(i). [8 marks]

(c) When **excess** lead trioxonitrate (V) solution was added to a solution of sodium chloride, the precipitate formed weighed 5.56 g.

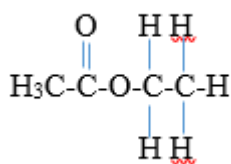
- Write a balanced equation for the reaction.
- Calculate the mass of sodium chloride in the solution. [Na = 23.0, Cl = 35.5, Pb = 207.0] [7 marks]

Question 33

- ☐ What is hydrogen electrode? [4 marks]
- ☐ Give a reason why metals are generally good reducing agents. [2 marks]
- ☐ State the reason why diamond is:
 - used in jewellery;
 - used as an abrasive;
 - a non-conductor of electricity. [3 marks]
- ☐ (i) What is a chemical change? (ii) Give two examples of a chemical change. [4 marks]
- ☐ What is an isotope? [2 marks]
- ☐ Differentiate between a weak acid and a dilute acid. [2 marks]
- ☐ Define the term pH. [2 marks]
- ☐ State two harmful effects that nitrogen (IV) oxide has on the environment. [2 marks]
- ☐ Write an equation for the reaction of chlorine with magnesium bromide. [2 marks]
- ☐ (i) State the method used to obtain pure ethanol from the aqueous mixture. (ii) State how the oxidation of ethanol can be prevented. [2 marks]

Question 34

- ☐ (i) Define the term functional group.
- (ii) Consider the following compound:



State the:

- I. functional group:
- II. name of the compound. [4 marks]

☐ (i) Name the class of organic compounds that could be identified using each of the following reagents:

- I. acidified tetraoxomanganate (VII) solution;
- II. ethanoic acid and conc. tetraoxosulphate (VI) acid;
- III. sodium trioxocarbonate (IV) solution.

(ii) State what is observed in each of the tests in 2(b)(i). [6 marks]

☐ (i) Consider the following reaction equation:



State the oxidation number of chlorine:

- I. before the reaction;
- II. after the reaction.

(ii) A ring is to be gold-plated by electrolysis using $\text{Au}(\text{NO}_3)_3$ solution.

- I. Write the reaction at the cathode.
- II. Calculate the time required to deposit 0.50 g gold on the ring when a current of 0.8 A is passed through the electrolyte.

$\text{Au} = 197, 1\text{F} = 96500 \text{ C}$ $\text{Au} = 197, 1\text{F} = 96500 \text{ C}$ [10 marks]

☐ Consider an ion with the symbol X^{2+} formed from its elemental state.

- (i) Which group does X belong in the periodic table?
- (ii) What is the formula of the compound formed from X^{2+} and a nitride ion?
- (iii) If X is in period 2, identify X.

Question 35

☐ Explain each of the following observations:

- (i) Sodium metal conducts electricity both in the solid and molten states, but sodium chloride only conducts electricity when molten.
- (ii) At room temperature, most covalent compounds are gases or liquids, whereas ionic compounds are solids. [8 marks]

- ☐ Give the IUPAC name of each of the following compounds:
- (i) $\text{CH}_3\text{C}(\text{CH}_3)_2\text{OH}$
 - (ii) $\text{C}(\text{CH}_3)_4$
 - (iii) HCOOH [3 marks]
- ☐ Identify the functional group in each of the compounds:
- (i) $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$
 - (ii) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
 - (iii) HCOOH [3 marks]
- ☐ When a given mass of CaCO_3 was heated, 0.25 dm^3 of a gas is collected at 25°C and at a pressure of 120 kNm^{-2} .
- (i) Write an equation for the reaction.
 - (ii) Calculate the mass of the:
 - (I) gaseous product obtained;
 - (II) CaCO_3 heated.

$\text{C}=12.0, \text{O}=16.0, \text{Ca}=40.0, \text{R}=8.314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$ $\text{C} = 12.0, \text{O} = 16.0, \text{Ca} = 40.0, \text{R} = 8.314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$ $\text{C}=12.0, \text{O}=16.0, \text{Ca}=40.0, \text{R}=8.314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$ [11 marks]

Question 36

- ☐ **(i) State the law of conservation of matter.**
- (ii) State three postulates of the kinetic theory of gases. [5 marks]
- ☐ **(b) Name the fuel produced when:**
- (i) Air is passed over heated carbon;
 - (ii) Steam is passed over heated carbon;
 - (iii) Write a balanced chemical equation for the reaction in 4(b)(ii). [4 marks]
- ☐ **(i) Name a suitable solvent for each of the following substances:**
- I. Sulphur;
 - II. Carbon(II) oxide;
 - III. Paint.
 - (ii) I. State what would be observed when dry hydrogen gas is passed over heated copper (II) oxide.
 - II. Write a chemical equation for the reaction in 4(c)(ii)I. [7 marks]
- ☐ **(i) Define air pollution.**
- (ii) State two methods of reducing carbon (II) oxide in the atmosphere. [4 marks]
- ☐ **(i) State three applications of electrolysis.**
- (ii) State two materials used in the extraction of iron from its ore. [6 marks]

Question 37

- a **(i) Name three gases that should be prepared in a fume chamber.**

- (ii) Give a reason for your answer in 5(a)(i).
- (iii) State one use of fused calcium chloride. [5 marks]

b (i) Describe briefly the laboratory preparation of dry ammonia gas.

- (ii) Explain briefly why ammonia cannot be dried with concentrated tetraoxosulphate (VI) acid. [5 marks]

c (i) Name the gas produced during the action of warm dilute HCl on:

- I. FeS;
- II. A mixture of Fe and S.
- (ii) Describe one chemical test to identify the gas produced in 5(c)(i) II.
- (iii) Give the reason why:
 - I. CuSO₄ solution is added to Zn and HCl during the preparation of hydrogen gas;
 - II. Chlorine is used for water treatment. [6 marks]

D: On analysis, 1.0 g sample of a hydrocarbon was found to contain 0.923 g of carbon. If the vapour density of the hydrocarbon is 39.0:

- (i) Determine its molecular formula;
- (ii) Name the compound. [H = 1.0; C = 12.0] [6 marks]

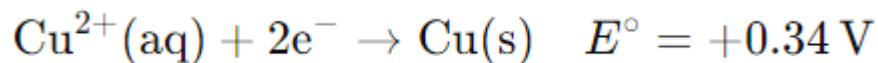
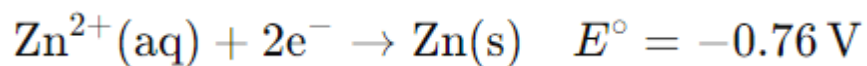
Question 38

- Define the term compound. [2 marks]
- State two conditions necessary for the cracking of petroleum fractions. [2 marks]
- Name two transition elements that are used as catalysts. [2 marks]
- Write an equation for the reaction between zinc dust and trioxonitrate (V) solution.
 - Which of the reactants in 1(d)(i) is:
 - I. Reduced;
 - II. Oxidized. [4 marks]
- Two isotopes of oxygen, ¹⁶O and ¹⁸O, have relative abundances of 90% and 10% respectively. Calculate the relative atomic mass of oxygen. [3 marks]
- List two ores of iron. [2 marks]
- What is biotechnology?
 - Name one product that can be obtained using biotechnology. [3 marks]
- Define the term element. [2 marks]
- State two sources of methane in the atmosphere. [2 marks]
- Explain briefly why the trend of the boiling points for Group VII elements is in the order I₂ > Br₂ > Cl₂ [3 marks]

Question 39

(a) Explain the statement, "the standard electrode potential of zinc is -0.76 V." What is meant by the term *periodic property of elements*?

(b) Consider the following standard electrode potentials:



When the **two** half cells are connected:

- (i) Write the reaction equation at each electrode.
- (ii) Write the overall cell reaction equation.
- (iii) State the type of reaction occurring at each electrode.
- (iv) Calculate the e.m.f. of the cell. [10 marks]

(c)

- (i) Name **two** chemical industries.
- (ii) State **two** factors that should be considered when siting a chemical industry.
- (iii) List **two** effects of a chemical industry on the community in which it is sited. [6 marks]

(d) Using chemical equations, explain *briefly* what would happen when hydrogen peroxide is added to:

- (i) Silver oxide.
- (ii) Chlorine gas. [6 marks]

(e) List three physical properties of nitrogen. [3 marks]

Question 40

(a) (i) What are acidic oxides?

(ii) Give one example of each of the following oxides:

I. acidic oxide;

II. basic oxide;

(III) amphoteric oxide;

(III) neutral oxide. [5 marks]

(b) (i) Define each of the following terms:

(I) Heat of combustion;

(II) Heat of neutralization.

(ii) Write an equation to illustrate each of the terms in 7(b)(i) above.

(iii) Given that the standard heat of combustion of butane (C_4H_{10}) is $+ 5877 \text{ kJmol}^{-1}$, calculate the heat of combustion of 14.5 g butane. [H = 1, C = 12] [10 marks]

(c) (i) Name two allotropes of sulphur.

(ii) State one difference between the two allotropes. [3 marks]

(d) (i) Give two characteristics of noble gases.

(ii) State one use each of

(I) He;

(II) Ar. [4 marks]

(e) State what is observed on warming ammonium trioxonitrate (V) with sodiumhydroxide. [2 marks]

Question 41

(a) (i) Define covalent bond.

(ii) Give two properties of covalent compounds

(iii) With the aid of a diagram, show how ammonia molecule is formed.

(iv) Illustrate with a diagram the formation of ammonium ion.

(v) What type of bond(s) exist(s) in

ammonia,

ammonium ion? [1H, 7 N] [10 marks]

(b) (i) Write three subatomic particles with their corresponding relative masses.

(ii) Name the possible states in which water can exist. [9 marks]

(c) (i) State Graham's law of diffusion.

(ii) Arrange the following gases, He, CH_4 and N_2 in order of increasing rates of diffusion. Give a reason for the order.[H = 1, He = 4, C = 12, N = 14] [4 marks]

(d) Draw the structures of the following compounds:

(i) 2, 3-dimethylbutane;

(ii) 1, 4-dibromocyclohexane. [2 marks]

Question 42

(a) Consider the following reaction sequence.

(i) What process leads to the formation of K?

(ii) Write the formula of K.

(iii) Write the structural formula of L and name L.

(iv) Name An.

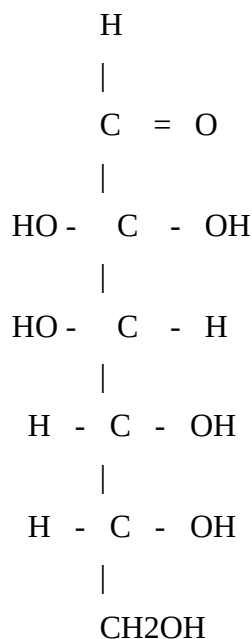
(v) Write the structure of M and name M. [7 marks]

(b) (i) What are carbohydrates?

(ii) Give one example each of a
Monosaccharide, disaccharide.

polysaccharide. [5 marks]

(c) Consider the following structure of a simple sugar.



(i) Which functional group makes the compound a reducing agent?

(ii) State what would be observed when

(I) the compound is mixed with Fehling's solution and boiled;

(II) few drops of concentrated H_2SO_4 is added to the sample of the compound.

(iii) Write an equation for the reaction in 3(c)(ii)(II). [6marks]

(d) A hydrocarbon Z with molecular mass 78 on combustion gave 3.385 g of CO_2 and 0.692 g of H_2O . Determine the molecular formula of Z.

[H = 1, C = 12, O = 16] [7 marks]

Question 44

(a) (i) Define in terms of electron transfer

I. oxidizing agent;

II. reducing agent.

(ii) Write a balanced equation to show that carbon is a reducing agent.

(iii) State the change in oxidation number of the specie that reacted with carbon in 2 (a)(ii). [5 marks]

(b) A gas X has a vapour density of 32. It reacts with sodium hydroxide solution to form salt and water only. It decolourizes acidified potassium tetraoxomanganate (VII) solution and reacts with H_2S to form sulphur. Using the information provided:

identify gas X;

state two properties exhibited by X;

give two uses of X. [5 marks]

(c) Consider the following substances:

(i) sodium;

(ii) lead (II) iodide;

(iii) hydrogen;

(iv) magnesium;

(v) oxygen.

Which of the substances

(i) conducts electricity?

(ii) is produced at the cathode during electrolysis of $\text{H}_2\text{SO}_4(\text{aq})$?

(iii) corresponds to the molecular formula AB_2 ?

(iv) is an alkaline earth metal? [5 marks]

(d) (i) Define the term salt.

(ii) Mention two types of salt.

(iii) Give an example of each of the salts mentioned in 2(d)(ii) above. [6 marks]

(e) In a neutralization reaction, dilute tetraoxosulphate (VI) acid completely reacted with sodium hydroxide solution.

(i) Write a balanced equation for the reaction.

(ii) How many moles of sodium hydroxide would be required for the complete neutralization of 0.50 moles of tetraoxosulphate (VI) acid? [4 marks]

Question 45

(a) State

(i) Pauli's Exclusion principle;

(ii) Hund's rule of maximum multiplicity. [4 marks]

(b) Write the electronic configuration of each of the following ions of copper:

I. Cu^+ ;

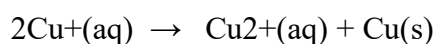
II. Cu^{2+} .

[29Cu]

(ii) Give the number of unpaired electrons in each of the ions in

1(b) (i) above.

(iii) State the type of reaction represented by the following equation:



(iv) Write the formula of one compound of Cu^{+} . [6 marks]

(c) (i) Name the type of radiation that will

- I. penetrate lead block;
- II. be stopped by thin paper.

(ii) Give the charge on each of the radiations mentioned in

1(c) (i) above.

(iii) What term is used to describe each of the following nuclear processes?

- I. Combination of two lighter nuclei to form a heavy nucleus
- II. Splitting of a heavy nucleus into two or more lighter nuclei
- III. Time required for one-half of the atoms of a radioactive substance to decay [7marks]

(d) Arrange the following ions in order of increasing size. Give a reason for your answer in each case.

- I. Li^{+} , K^{+} , Na^{+} ;
- II. O^{2-} , F^{-} , N^{3-} . [4 marks]

(e) Determine the percentage composition of phosphorus and oxygen in phosphorus (V) oxide.

[P = 31, O = 16] [4 marks]

Question 46

(a) What is a transition element? [2 marks]

(b) Iron can be represented as ^{56}Fe .

- 1. Write the electron configuration for iron.
- 2. Explain briefly why iron exhibits:
 - I. Paramagnetism;
 - II. Variable oxidation.
- 3. Mention two by-products in the extraction of iron in the blast furnace.
- 4. Write an equation to illustrate the formation of each of the by-products in 1(b)(iii).

[10 marks]

(c) List the elements present in each of the following alloys:

- 1. Steel;
- 2. Bronze;
- 3. Brass;
- 4. Soft solder. [4 marks]

(d) Outline how a pure dry sample of silver chloride could be obtained from aqueous solutions of sodium chloride and silver trioxonitrate (V). [5 marks]

(e) If 2.40 g of carbon is burnt completely in air, calculate the volume of carbon (IV) oxide produced at s.t.p.

[C=12.0, Molar volume, $V_m=22.4 \text{ dm}^3\text{mol}^{-1}$][C = 12.0 1]

[4 marks]

Question 46

(a) Consider the following list of elements: nitrogen, fluorine, aluminium, and potassium.

Which of the elements:

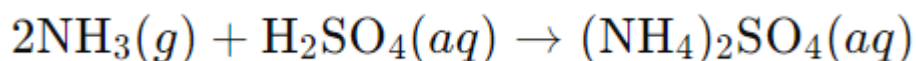
1. Forms a diatomic molecule with a triple bond?
2. Is the most reactive electropositive element?
3. Forms amphoteric oxide?
4. Is the most reactive non-metal? **[4 marks]**

(b)

1. State what would be observed if chlorine gas is passed into an aqueous solution of:
 - I. Iron (II) chloride;
 - II. Potassium bromide.
2. Give the names of the products in 2(b)(i)I and 2(b)(i)II.
3. Write an ionic equation to represent each of 2(b)(i)I and 2(b)(i)II.
4. Suggest the type of reaction in each of 2(b)(i)I and 2(b)(i)II. **[13 marks]**

(c) State two differences between conductors and electrolytes. **[2 marks]**

(d) Consider the reaction represented by the following chemical equation:



Determine the mass of ammonium tetraoxosulphate (VI) that would be produced from 85 g of ammonia.

[H = 1.00, N = 14.0, O = 16.0, S = 32.0]

[6 marks]

Question 47

(a)

1. What is the shape of:
 - I. s-orbital;
 - II. p-orbital?
2. Which ion has the following composition?

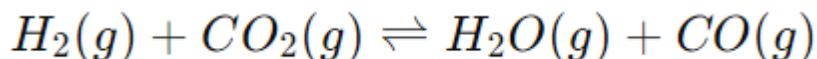
- 0 electron, 2 protons, and 2 neutrons. **[3 marks]**

(b) The following table shows the melting and boiling points of oxides of the elements XXX, YYY, and ZZZ.

Oxides	Melting point/K	Boiling point/K
X ₂ OX ₂ O	1403	2223
Y ₂ OY ₂ O	273	373
Z ₂ OZ ₂ O	49	128

1. What type of bond binds XXX, YYY, and ZZZ respectively to oxygen in their oxides?
2. Explain briefly how the bond in Z₂OZ₂O is formed?
3. What type of forces hold the molecules of:
 - I. Y₂OY₂O
 - II. Z₂OZ₂O respectively? **[8 marks]**

(c) The following equation illustrates homogeneous equilibrium established when hydrogen and carbon (IV) oxide react:



1. Why is this reaction regarded as homogeneous?
2. Explain briefly the effect of an increase in pressure on the:
 - I. Equilibrium position;
 - II. Reaction rate.
3. State three features of an equilibrium reaction. **[8 marks]**

(d) A compound Q contains 29.1% sodium, 40.5% sulfur, and 30.4% oxygen. Determine the:

1. Empirical formula;
2. Molecular formula of Q if its relative molar mass is 158.

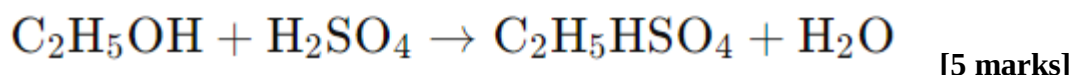
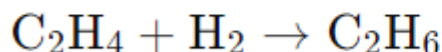
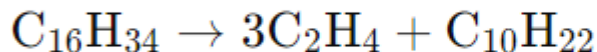
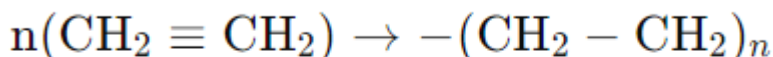
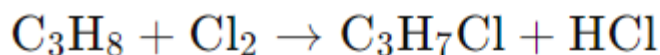
$$[O = 16.0, Na = 23.0, S = 32.0] \quad \textbf{[6 marks]}$$

Question 48

(a)What is meant by saponification?

- i. List the raw materials needed for the manufacture of soap.
- ii. Name the main by-product obtained from the manufacture of soap. **[5 marks]**

(b)



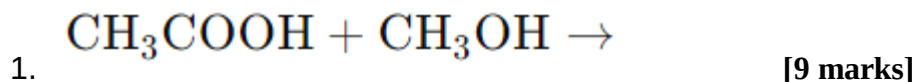
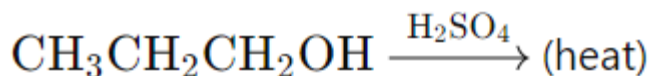
(c) An organic compound U produces effervescence when reacted with sodium metal, liberating gas V. U produces a sweet fruity smelling liquid when warmed with colorless liquid W in the presence of a catalyst. W reacts with sodium hydrogen trioxocarbonate (IV) solution to produce gas G.

1. Write the functional group present in U and W.
2. Identify gases V and G.
3. Name the type of reaction between:
 - o I. U and sodium metal;
 - o II. U and W.
4. Name the catalyst used in the reaction between U and W. [5 marks]

Question 49

(a) What is fermentation?

1. Write an equation for the fermentation of glucose.
2. What substance must be added to glucose solution to make it ferment?
3. Complete and balance the following chemical equations:

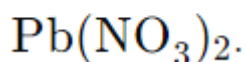
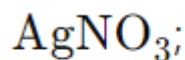


(b) Explain why ethanol is completely miscible with water. [3 marks]

(c)

1. What is biotechnology?
2. Distinguish between fine chemicals and heavy chemicals.
3. Mention the major raw material used in the large-scale production of each of the following products:
 1. cement;
 2. tyre. [6 marks]

- (d) i. List three chemical properties of acids.
- ii. Give two large-scale uses of H_2SO_4 .
- iii. Write a chemical equation for the action of heat on each of the following compounds:



[7 marks]

Question 50

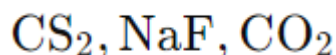
- Define hybridization.
 - Explain briefly how sp^3 hybrid orbitals are formed. [5 marks]
- (b) Consider the equilibrium reaction represented by the following equation:
- $$2\text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g); \Delta H = -x \text{ kJ mol}^{-1}$$
- Write an expression for the equilibrium constant K_p .
 - Sketch the energy profile diagram for the forward reaction indicating the position of:
 - reactants and products;
 - activated complex;
 - enthalpy change;
 - activation energy. [7 marks]

Question 50

(a) Describe briefly how each of the following aqueous solutions could be identified in the laboratory:

- ammonium trioxocarbonate (IV);
- ammonium chloride. [6 marks]

(b) Arrange the following compounds in order of increasing boiling point and give reasons



for your answer [3 marks]

(c) List two gases each that are:

- acidic;
- highly soluble in water;
- oxidized by acidified KMnO_4 . [6 marks]

(d) In a tabular form, compare the elements silicon and sulphur under the following properties:

- metallic character;
- physical state;
- conduction of electricity. [3 marks]

(e)

(e) A cuboid piece of sodium metal measures $3\text{ cm} \times 4\text{ cm} \times 10\text{ cm}$. If the density of sodium is 0.971 g cm^{-3} , calculate the number of atoms in the sodium metal.

[Na = 23; Avogadro constant = $6.02 \times 10^{23}\text{ mol}^{-1}$]

[5 marks]